

Project 0: Getting Real

Preliminaries

Fill in your name and email address.

钟骏宇, 2400012946@stu.pku.edu.cn

If you have any preliminary comments on your submission, notes for the TAs, please give them here.

Please cite any offline or online sources you consulted while preparing your submission, other than the Pintos documentation, course text, lecture notes, and course staff.

Booting Pintos

A1: Put the screenshot of Pintos running example here.

```
KUOS/pintos/src/threads/build'
root@6888fe2d00c9:~/pintos/src/threads# cd build
root@6888fe2d00c9:~/pintos/src/threads/build# pintos --
qemu-system-i386 -device isa-debug-exit -drive format=raw,media=disk,index=0,file=/tmp/hGo08PE20R.dsk -m 4 -net none -
nographic -monitor null
Pintos hda1
Loading.....
Kernel command line:
Pintos booting with 3,968 kB RAM...
367 pages available in kernel pool.
367 pages available in user pool.
Calibrating timer... 157,081,600 loops/s.
Boot complete.

KUOS/pintos/src/threads/build'
squish-pty bochs -q
=====
Bochs x86 Emulator 2.6.2
Built from SVN snapshot on May 26, 2013
Compiled on Mar 1 2022 at 16:09:16
=====
0000000000i[ ] reading configuration from bochsrc.txt
0000000000e[ ] bochsrc.txt:8: 'user_shortcut' will be replaced by new 'keyboard' option.
0000000000i[ ] installing nogui module as the Bochs GUI
0000000000i[ ] using log file bochsout.txt
Pintos hda1
Loading.....
Kernel command line:
Pintos booting with 4,096 kB RAM...
383 pages available in kernel pool.
383 pages available in user pool.
Calibrating timer... 102,400 loops/s.
Boot complete.
```

Debugging

QUESTIONS: BIOS

B1: What is the first instruction that gets executed?

```
1 jmp $0x3630,$0xf000e05b
```

B2: At which physical address is this instruction located?

0xFFFFF0

QUESTIONS: BOOTLOADER

B3: How does the bootloader read disk sectors? In particular, what BIOS interrupt is used?

via int 0x13, specifically in `read_sector` it sets `AH = 0x42` and invokes int 0x13 (int13 Extended Read Sectors)

B4: How does the bootloader decide whether it successfully finds the Pintos kernel?

The bootloader scans the partition table on each system hard disk, and check whether it is unused, a Pintos kernel partition (type is 0x20) and a bootable partition (type is 0x80). When it finds such a partition, it assumes that the Pintos kernel is there and starts loading it.

B5: What happens when the bootloader could not find the Pintos kernel?

the bootloader prints `"\r\nNot found\r\n"` and invokes `int 0x18` to report boot failure.

B6: At what point and how exactly does the bootloader transfer control to the Pintos kernel?

at 0x7cbf. After loading the kernel into memory, the bootloader reads the kernel entry point from the loaded ELF header, constructs a real-mode far pointer, and transfers control with `ljmp *start`

QUESTIONS: KERNEL

B7: At the entry of `pintos_init()`, what is the value of expression

`init_page_dir[pd_no(ptov(0))]` in hexadecimal format?

Ans: 0x0

B8: When `palloc_get_page()` is called for the first time,

B8.1 what does the call stack look like?

Ans:

#0 `palloc_get_page` (flags=PAL_ASSERT | PAL_ZERO) at `../threads/palloc.c:113`

#1 `0xc00203aa` in `paging_init ()` at `../threads/init.c:168`

#2 `0xc002031b` in `pintos_init ()` at `../threads/init.c:100`

#3 `0xc002013d` in `start ()` at `../threads/start.S:180`

B8.2 what is the return value in hexadecimal format?

Ans: 0xc0101000

B8.3 what is the value of expression `init_page_dir[pd_no(ptov(0))]` in hexadecimal format?

Ans: 0x0

B9: When `palloc_get_page()` is called for the third time,

B9.1 what does the call stack look like?

Ans:

#0 `palloc_get_page` (flags=PAL_ZERO) at `../threads/palloc.c:113`

#1 `0xc0020a81` in `thread_create` (name=0xc002e895 "idle", priority=0, function=0xc0020eb0, aux=0xc000efbc) at `../threads/thread.c:178`

#2 `0xc0020976` in `thread_start ()` at `../threads/thread.c:111`

#3 `0xc0020334` in `pintos_init ()` at `../threads/init.c:119`

#4 `0xc002013d` in `start ()` at `../threads/start.S:180`

B9.2 what is the return value in hexadecimal format?

Ans:0xc0103000

B9.3 what is the value of expression `init_page_dir[pd_no(ptov(0))]` in hexadecimal format?

Ans:0x102027

Kernel Monitor

C1: Put the screenshot of your kernel monitor running example here. (It should show how your kernel shell respond to `whoami`, `exit`, and other input.)

```
PS G:\Workspace\Code\2Down\OS> .\pintos-up.ps1
2400012946
PKUOS>exit
qemu-system-i386: terminating on signal 2

root@d4652a8c37c4:~/pintos/src/threads/build# pintos --
qemu-system-i386 -device isa-debug-exit -drive format=raw,media=disk,index=0,file=/tmp/SH7D_BwsNy.dsk -m 4 -net none -nographic -monitor
null
Pintos hda1
Loading.....
Kernel command line:
Pintos booting with 3,968 kB RAM...
367 pages available in kernel pool.
367 pages available in user pool.
Calibrating timer... 209,510,400 loops/s.
Boot complete.
PKUOS>whoami
2400012946
PKUOS>aba
invalid command
PKUOS>exit
```

####

C2: Explain how you read and write to the console for the kernel monitor.

Use the functions in `input.c` and `printf()`.

"`Read_line_myself()`" realize the input and echo and check whether it is `'\r'('\n')`, printable symbol or none of them, more than that the function can handle "backspace" key.

and do some if and else in the "`pintos_init()`" to handle command "`whoami`" and "`exit`", just print or break.

It is also important to note that `input_init()` has already been executed during kernel initialization, as a part of my work of reading program:)